

DIGITAL WALLET SYSTEM

Secure Online Payment & Fund Transfer Platform

1. Project Overview

The Digital Wallet System is a secure web-based financial application built using Spring Boot (Java), React, MySQL, Swagger, and deployed on AWS Cloud. It allows users to register, login, add money, send money, and view transaction history securely.

2. Problem Statement

Traditional payment systems face challenges such as security risks, complex banking procedures, and lack of real-time tracking. A secure and scalable digital wallet is required for seamless transactions.

3. Solution

A full-stack web application providing secure authentication (JWT), wallet management, peer-to-peer transfers, transaction tracking, and cloud deployment.

4. MVP Features

• User Registration & Login • Add Money to Wallet • Send Money to Registered Users • View Transaction History • Swagger API Documentation • AWS Deployment

5. Technology Stack

Backend: Java, Spring Boot, Spring Security, JWT Frontend: React JS, Axios Database: MySQL
Cloud: AWS EC2, AWS RDS, AWS S3

6. User Flow

Register → Login → Add Money → Send Money → View Transactions → Logout

7. Database Design

Users Table, Wallet Table, Transactions Table with proper relationships and constraints.

8. Security Implementation

• Password Encryption (BCrypt) • JWT Token Authentication • Role-Based Authorization • Secure REST APIs

9. Success Metrics

• Transaction success rate • User growth • System uptime • API performance
